AQI Prediction — Indian Cities Report

Prepared By – Ayeshkant Ray

Course – SparkIIT Data Science

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Tools – Python (numpy, pandas, seaborn, matplotlib, sklearn) Power BI

1. Abstract

This report presents a machine learning approach to predicting the Air Quality Index (AQI) across Indian cities using city-level daily pollutant concentration readings. The dataset contains measurements of twelve pollutants - including PM2.5, PM10, NO2, SO2, CO, O3, NH3, Benzene, Toluene, and Xylene - alongside corresponding AQI values, aggregated by city and date.

Two regression models were trained and compared: Linear Regression as a baseline, and Random Forest Regressor as a more expressive ensemble model. A random 80/20 train-test split was used — appropriate for this dataset because the readings span multiple independent cities rather than forming a single continuous time series.

According to exploratory data analysis, North Indian cities (Delhi, Patna, Ahmedabad) consistently record the highest average AQI values; temperature inversions and crop stubble burning cause sharp pollution spikes during the winter months (October–January); and PM2.5 is the primary predictor of AQI, according to both correlation analysis and Random Forest feature importance rankings. Given the nearly formulaic association between contaminants and AQI, linear regression performed well. By capturing non-linear interactions between pollutants, Random Forest significantly enhanced. City-level time-series forecasting with ARIMA or Prophet is the best suggestion for further research.

The project is further supported by an interactive 4-page Power BI dashboard covering Air Quality Overview, Seasonal & Temporal Trends, Pollutant Relationships & Model Insights, and Model Performance & Evaluation. Dashboard analysis across 26 cities and 24,850 records confirms an average AQI of 166.46, a peak of 2,000, and 1,000 total severe days across the dataset period.

2. Introduction

In India, one of the most urgent public health issues is air quality.
Indian cities frequently rank among the most polluted in the world, having detri mental effects on respiratory health, life expectancy, and economic output, accor ding to numerous yearly worldwide air quality studies.
The Central Pollution Control Board (CPCB) of India uses the Air Quality Inde x (AQI) as a standard metric to inform the public about daily air quality levels o n a uniform 0–500 scale, where higher values indicate worse air quality.

Sub-indices of individual pollutants are used to create the AQI; each sub-index is derived from the measured concentration in relation to national ambient air quality requirements, and the maximum sub-index is used to determine the overall AQI. This indicates that AQI is a well-defined regression target with a strong feature-target signal since it has an intrinsically organized mathematical connection with pollutant concentrations.

In order to learn and forecast this relationship from past data, our study use machine learning. Although the deterministic formula is known, pure formula application is unable to account for noise, measurement gaps, and city-specific patterns found in real-world sensor data. Additionally, a data-driven model can identify the contaminants that are most predictive under real Indian urban situations, which is a topic of interest to both scientists and policymakers.

Two models are compared: Linear Regression, which tests whether the pollutant-AQI relationship is well-approximated linearly; and Random Forest Regressor, which can capture interaction effects between pollutants that a linear model cannot represent.

The analysis is complemented by an interactive Power BI dashboard that provides exploratory visual analysis across four pages, enabling cross-filtering by City, AQI Category, Month, and Year.

3. Objectives

The primary objectives of this project are as follows:

- To explore the AQI dataset through visualisations of the AQI distribution, city-level pollution rankings, seasonal trends over time, and pollutant correlation structure - presented through both Python plots and an interactive Power BI dashboard.
- To handle missing values in pollutant readings appropriately — using column-wise median imputation to account for the right-skewed distributions caused by extreme pollution events.
- To engineer temporal features (Month, Year) from the Date column to allow models to learn seasonal pollution patterns.
- To identify and drop leakage-causing features (AQI_Bucket) and non-predictive identifiers (City, Date) before modelling.
- To train a Linear Regression baseline and evaluate it using R^2 Score, RMSE, Actual vs Predicted plot, and Residual plot.
- To train a Random Forest Regressor and compare its performance against the linear baseline using the same evaluation metrics.
- To extract and interpret Random Forest feature importance scores, validating them against the correlation analysis from EDA.
- To draw conclusions about model suitability for AQI prediction and outline a roadmap for city-level temporal forecasting.

4. Literature Review / Existing Work

The Air Quality Index (AQI) systems, established by the US EPA in the 1970s and adapted in India in 2014, standardize air pollution communication impacting public health. AQI, calculated deterministically from sub-indices, contrasts traditional machine learning regression by providing a more structured approach. Early prediction models, like ARIMA, effectively captured temporal patterns but were limited by their univariate nature. The advent of machine learning methods has allowed for better predictions using regression models such as Linear Regression and Support Vector Regression, although they often underestimate AQI during extreme events.

Ensemble methods like Random Forest and XGBoost have emerged to tackle these challenges by capturing non-linear interactions. Deep learning approaches, particularly Long Short-Term Memory networks, show potential for multi-step forecasting. However, the Indian context presents unique challenges, with varied pollution sources affecting urban centers differently and seasonal agricultural practices causing localized spikes in AQI. Despite model advancements, issues like sparse sensor data and static models failing to adapt to changes underscore the need for enhanced, city-specific approaches that incorporate relevant meteorological factors.

5. Dataset Description

The dataset used is a single CSV file (city_day.csv) containing city-level daily air quality readings for multiple Indian cities. Each row represents one city's pollutant measurements on a given date, along with the computed AQI value.

5.1 Key Columns

- City — name of the Indian city (dropped at modelling stage as a non-predictive identifier)
- Date — calendar date in YYYY-MM-DD format (used for temporal feature extraction, then dropped)
- AQI — Air Quality Index value (target variable; rows with null AQI are removed)
- AQI_Bucket — categorical label derived from AQI (e.g. 'Good', 'Moderate', 'Hazardous') — dropped to prevent data leakage

5.2 Pollutants Features

Pollutant	**Correlation with AQI**	**Role in Air Quality**
PM2.5	Very High	Fine particulate matter — primary AQI driver in India; penetrates deep into lungs
PM10	High	Coarse particles — construction dust, road dust; strong AQI contributor
NO / NO2 / NOx	Moderate	Vehicle and industrial emissions; contribute to smog and respiratory issues
NH3	Moderate	Ammonia from agriculture and fertilisers; reacts to form secondary particulates
SO2	Low–Moderate	Industrial combustion; forms sulphate aerosols
CO	Low	Vehicle exhaust; weaker direct AQI correlation in this dataset
O3	Low	Ground-level ozone; formed from NOx + sunlight reactions
Benzene / Toluene / Xylene	Weak	Volatile organic compounds; indirect AQI influence

5.3 Missing Data

The dataset has significant missing values — not all pollutants are measured every day in every city. AQI itself is also missing for some dates. Rows with null AQI were removed entirely (as they cannot serve as training targets), and missing pollutant readings were filled with column-wise medians to preserve as many usable rows as possible.

6. Methodology

The project follows a supervised machine learning pipeline across ten stages:

SNo.	Stage	Description
1	**Data Loading**	Single CSV (city_day.csv) with city-wise daily pollutant readings and AQI values
2	**Null Analysis**	Identified significant missing data — not all pollutants measured every day in every city
3	**AQI Filtering**	Dropped rows where AQI is null — target variable must be present for all training samples
4	**Null Imputation**	Filled missing pollutant values with column-wise median (robust to extreme pollution spikes)
5	**Feature Engineering**	Extracted Month and Year from Date to capture seasonal and annual pollution trends
6	**Feature Selection**	Dropped AQI_Bucket (data leakage), City and Date (identifiers, not features)
7	**Train/Test Split**	Random 80/20 split — appropriate for city-aggregated, non-time-series data
8	**Model 1 — LR**	Linear Regression baseline; evaluated with RMSE, R^2, Actual vs Predicted, Residual plot
9	**Model 2 — RF**	Random Forest (100 trees); evaluated on same metrics; feature importance extracted
10	**Comparison**	Side-by-side RMSE and R^2 comparison to quantify the improvement from LR to RF

6.1 Data Loading

The dataset was loaded from a single CSV file using pandas. Initial inspection via df.head(), df.info(), and df.isnull().sum() revealed the shape, data types, and the extent of missing values across pollutant columns.

6.2 Null Value Handling

AQI rows with null values were dropped first — the target variable must be present for every training sample. For the twelve pollutant feature columns, missing values were filled with column-wise median rather than mean. The justification is that AQI data is right-skewed, with occasional extreme pollution spikes (e.g. during crop burning or industrial accidents) that pull the mean upward; the median is more representative of typical conditions for a given pollutant.

6.3 Feature Engineering

Month and Year were extracted from the Date column using pandas dt accessor. These features encode seasonal structure (winter pollution peaks) and year-on-year trends into the model without exposing the raw date string, which is non-numeric and cannot be used directly by scikit-learn estimators.

6.4 Feature Selection

AQI_Bucket was dropped because it is a categorical re-encoding of the target variable (AQI) — including it would cause severe data leakage, giving the model direct access to the answer. City and Date were dropped as non-predictive identifiers. The resulting feature set consists of the twelve pollutant columns plus Month and Year.

6.5 Train-Test Split

A random 80/20 split was used with random_state=42. This contrasts with the Rossmann project, which required a time-based split to prevent leakage of future sales into training. Here, the dataset aggregates independent readings across many cities — there is no single time series that would leak future dates into the past. A random split is therefore appropriate and maximises the diversity of training data.

6.6 Model 1 — Linear Regression

A Linear Regression model was instantiated with default scikit-learn hyperparameters and fitted on the training set. Because AQI is mathematically derived from pollutant concentrations via a structured formula, a linear model is expected to perform well — the underlying relationship is not arbitrary but is grounded in a known sub-index computation.

6.7 Model 2 — Random Forest Regressor

A Random Forest with 100 trees (n_estimators=100) was trained on the same features and split. Random Forest builds an ensemble of decision trees, each trained on a bootstrap sample of the data with a random subset of features. Final predictions are averaged across all trees, reducing overfitting and capturing non-linear interactions — for example, the combined effect of high PM2.5 and high

NO2 on AQI may be larger than their individual contributions, which a linear model cannot express.

6.8 Evaluation Framework

Both models were evaluated on the same held-out test set using R^2 Score (proportion of AQI variance explained) and RMSE (average prediction error in AQI units). Diagnostic plots — Actual vs Predicted and Residual — were generated for both models to assess prediction quality and identify systematic errors.

7. Exploratory Data Analysis — Key Findings

7.1 AQI Distribution

The AQI distribution across all cities and dates is right-skewed: the majority of readings fall in the moderate range (AQI 50–200), but a long tail of severely polluted days extends to values above 400. This skew reflects the episodic nature of extreme pollution events — crop burning, winter inversions, and industrial incidents that dramatically spike readings for short periods.

→ Right-skewed AQI distributions mean that standard regression metrics (RMSE) are heavily influenced by extreme pollution events. Models that underpredict these peaks will show high RMSE even if they predict typical days accurately.

7.2 Most Polluted Cities

Ranking cities by average AQI reveals a clear North India dominance at the top of the pollution table. Cities such as Delhi, Patna, and Ahmedabad consistently record the highest average AQI values, driven by a combination of vehicular density, industrial activity, proximity to agricultural burning zones in Punjab and Haryana, and unfavourable meteorology in winter months. Southern and coastal cities tend to record significantly lower average AQI.

→ City identity is a strong implicit predictor of AQI — but it was deliberately excluded from the feature set as a non-predictive identifier. A more sophisticated model could encode city as a fixed effect or use embeddings to capture location-specific pollution baselines.

7.3 AQI Trend Over Time

The time-series plot of average daily AQI across all cities shows a recurring seasonal pattern: AQI rises sharply from October through January and falls back in the warmer months. This pattern is consistent across years and is driven by temperature inversions in winter (cold, still air trapping pollutants near the ground), increased use of heating fuels, and the October–November crop stubble burning season in North India. A mild secondary peak sometimes appears around Diwali in October–November.

→ The Month feature engineered from the Date column directly encodes this seasonality, giving both models a way to learn winter-peak patterns without processing raw date strings.

7.4 Pollutant Correlation with AQI

The correlation heatmap of pollutant features against AQI reveals a clear hierarchy of predictive relevance:

- PM2.5 and PM10 — very high positive correlation with AQI. Consistent with India's AQI methodology where particulate matter sub-indices are dominant.
- NO, NO2, NOx — moderate positive correlation. Nitrogen oxides contribute significantly during high-traffic periods.
- NH3 — moderate correlation, driven by agricultural emission patterns in certain seasons.
- SO2 — low to moderate; more relevant near industrial clusters.
- CO, Benzene, Toluene, Xylene — weak direct correlation with AQI, though they contribute to overall air toxicity.
- O3 — ground-level ozone shows weak or sometimes negative correlation with AQI in this dataset, reflecting its photochemical formation pattern (higher in summer when AQI tends to be lower).

→ PM2.5 emerges as the single most correlated pollutant with AQI — a finding later confirmed by Random Forest feature importance scores, providing strong cross-validation of the EDA insight.

8. Code

8.1 Importing Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score
```

8.2 Loading the Dataset

```
df = pd.read_csv("../Datasets/AQI dataset/city_day.csv")
print(df.shape)
df.head()

df.info()

df.isnull().sum()
```

8.3 Null Value Handling

```
df = df.dropna(subset=["AQI"])
print("Shape after dropping null AQI:", df.shape)
### Step 2 — Fill Missing Pollutant Values
Filling missing pollutant readings with **column-wise median**
pollutant_cols = ["PM2.5", "PM10", "NO", "NO2", "NOx", "NH3", "CO", "SO2", "O3",
"Benzene", "Toluene", "Xylene"]
for col in pollutant_cols:
    df[col] = df[col].fillna(df[col].median())
```

8.4 EDA — AQI Distribution and City Rankings

```
# AQI Distribution

sns.histplot(df["AQI"].dropna(), bins=50, kde=True)
plt.title("AQI Distribution")
plt.xlabel("AQI")
plt.show()

# Top 10 polluted cities

data = df.groupby("City")["AQI"].mean().sort_values(ascending=False).head(10).reset_index()
sns.barplot(x="AQI", y="City", data=data)
plt.title("Top 10 Most Polluted Cities (Avg AQI)")
plt.show()
```

8.5 EDA — AQI Trend Over Time and Correlation Heatmap

```
# AQI Trend over time

df["Date"] = pd.to_datetime(df["Date"])
daily_aqi = df.groupby("Date")["AQI"].mean()
plt.figure(figsize=(14, 4))
plt.plot(daily_aqi)
plt.title("Average AQI Over Time (All Cities)")
plt.xlabel("Date")
plt.ylabel("AQI")
plt.show()

# Correlation Heatmap

pollutants = ["PM2.5", "PM10", "NO", "NO2", "NOx", "NH3", "CO", "SO2", "O3", "Benzene",
"Toluene", "Xylene", "AQI"]
plt.figure(figsize=(12, 8))
sns.heatmap(df[pollutants].corr(), annot=True, fmt=".2f", cmap="coolwarm")
plt.title("Pollutant Correlation with AQI")
plt.show()
```

8.6 Feature Engineering and Selection

```
df["Month"] = df["Date"].dt.month
df["Year"] = df["Date"].dt.year

# AQI_Bucket is derived from AQI, City and Date not needed for LR
df_model = df.drop(columns=["City", "Date", "AQI_Bucket"])
df_model.head()
X = df_model.drop(columns=["AQI"])
y = df_model["AQI"]
print("Features:", X.columns.tolist())
```

8.7 Train-Test Split

```
# Using random split here since this is city-level aggregated data, not a single time series
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
print("Train size:", X_train.shape, "| Test size:", X_test.shape)
```

8.8 Model 1 — Linear Regression

```
lr = LinearRegression()
lr.fit(X_train, y_train)
y_pred_lr = lr.predict(X_test)

rmse_lr = np.sqrt(mean_squared_error(y_test, y_pred_lr))
r2_lr = r2_score(y_test, y_pred_lr)
print(f"Linear Regression → RMSE: {rmse_lr:.2f} | R²: {r2_lr:.4f}")
```

```
# Actual vs Predicted Plot

plt.scatter(y_test, y_pred_lr, alpha=0.3)
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--')
plt.xlabel("Actual AQI")
plt.ylabel("Predicted AQI")
plt.title("LR: Actual vs Predicted AQI")
plt.show()

# Residual Plot

residuals = y_test- y_pred_lr
plt.scatter(y_pred_lr, residuals, alpha=0.3)
plt.axhline(0, color='red')
plt.xlabel("Predicted AQI")
plt.ylabel("Residuals")
plt.title("LR: Residual Plot")
plt.show()
```

8.9 Model 2 — Random Forest Regressor

```
rf = RandomForestRegressor(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)
y_pred_rf = rf.predict(X_test)

rmse_rf = np.sqrt(mean_squared_error(y_test, y_pred_rf))
r2_rf = r2_score(y_test, y_pred_rf)
print(f"Random Forest → RMSE: {rmse_rf:.2f} | R²: {r2_rf:.4f}")

# Actual vs Predicted Plot

plt.scatter(y_test, y_pred_rf, alpha=0.3)
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--')
plt.xlabel("Actual AQI")
plt.ylabel("Predicted AQI")
plt.title("RF: Actual vs Predicted AQI")
plt.show()
```

8.10 Feature Importance — Random Forest

```
importance = pd.Series(rf.feature_importances_,
index=X.columns).sort_values(ascending=False)
sns.barplot(x=importance.values, y=importance.index)
plt.title("Feature Importance (Random Forest)")
plt.xlabel("Importance Score")
plt.show()

print(importance)
```

8.11 Model Comparison

```
print(f"{'Model':<20} {'RMSE':>8} {'R²':>8}")
print(f"{'Linear Regression':<20} {rmse_lr:>8.2f} {r2_lr:>8.4f}")
print(f"{'Random Forest':<20} {rmse_rf:>8.2f} {r2_rf:>8.4f}")
```

9. Output Snapshots

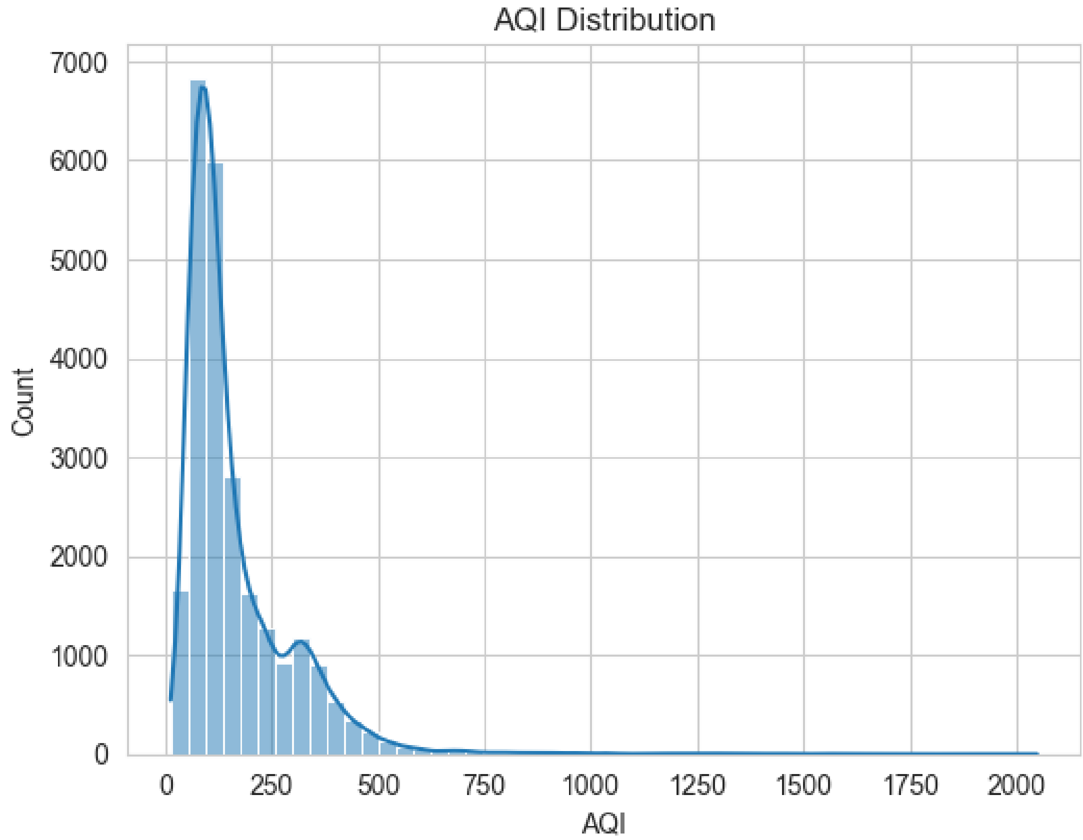


Figure 1

This histogram with a KDE (kernel density estimate) overlay displays the frequency distribution of AQI values across all cities and dates. The distribution is visibly right-skewed, with the bulk of readings concentrated in the moderate AQI range (50–200), and a long tail extending toward values above 400. The skew reflects the episodic nature of extreme pollution events — crop burning, winter inversions — that produce severe spikes for short periods while typical days remain in the moderate range.

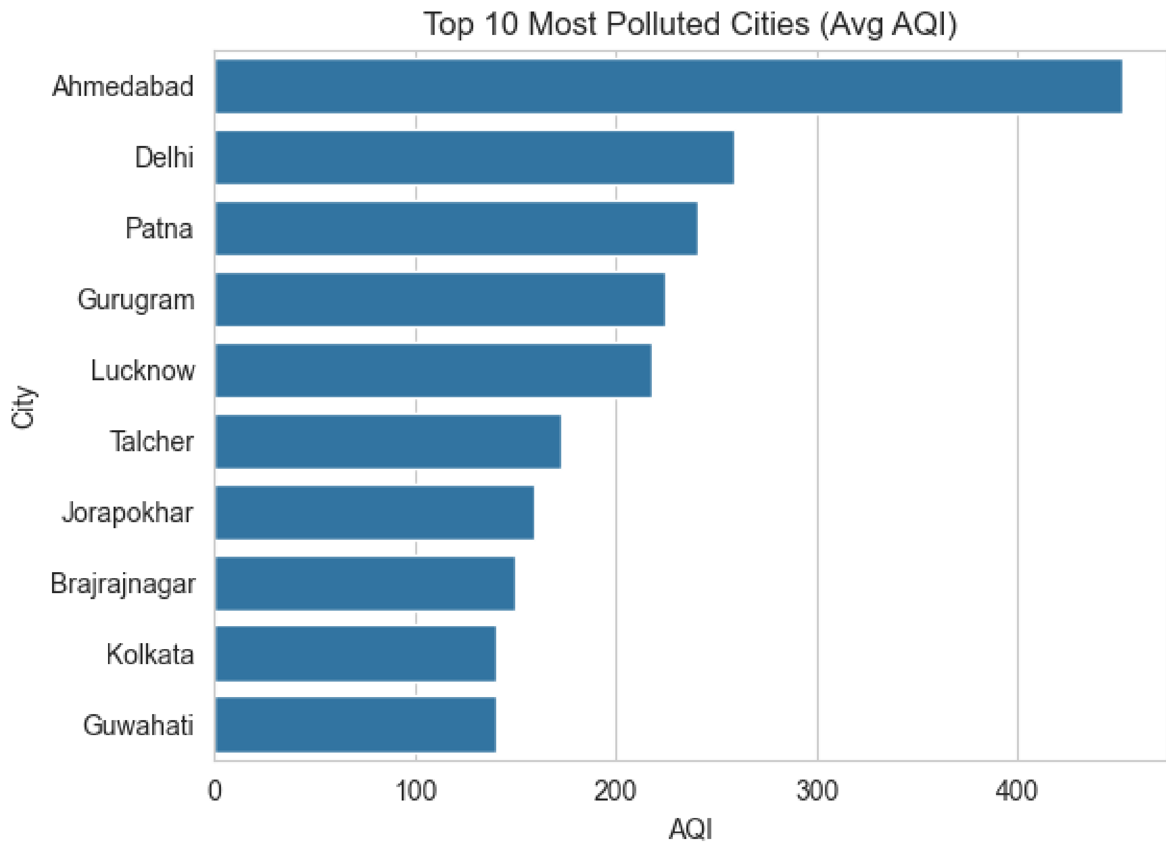


Figure 2

This horizontal bar chart ranks the ten most polluted cities by their mean AQI value across the dataset. North Indian cities dominate the top positions, with Delhi, Patna, and Ahmedabad recording the highest averages. The chart visually confirms the North India pollution dominance documented in environmental literature and validates the EDA finding that city identity is a strong implicit predictor of AQI.

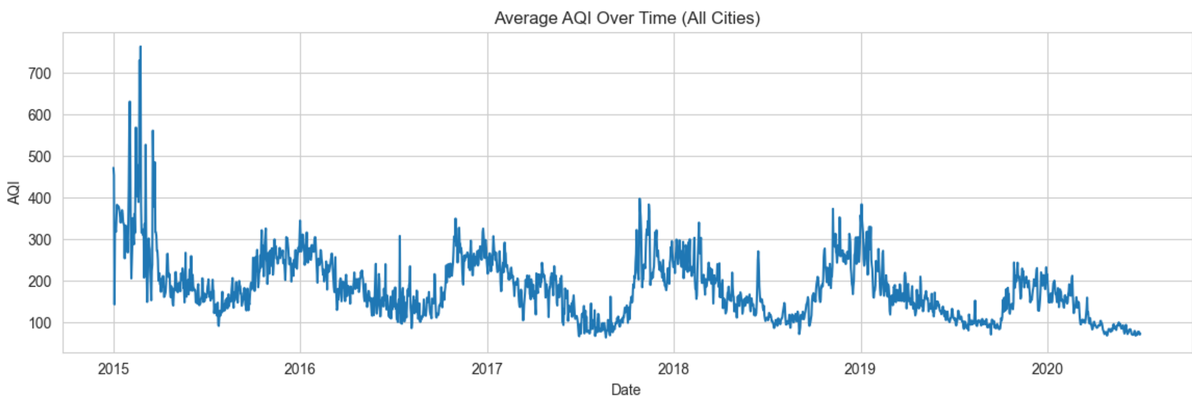


Figure 3

This time-series line plot shows the mean daily AQI aggregated across all cities from the start to the end of the dataset's date range. A clear recurring seasonal pattern is visible — AQI rises sharply in October–January and falls back through the warmer months. The winter peaks, driven by temperature inversions and stubble burning, appear consistently across years, validating the importance of the Month feature engineered for modelling.

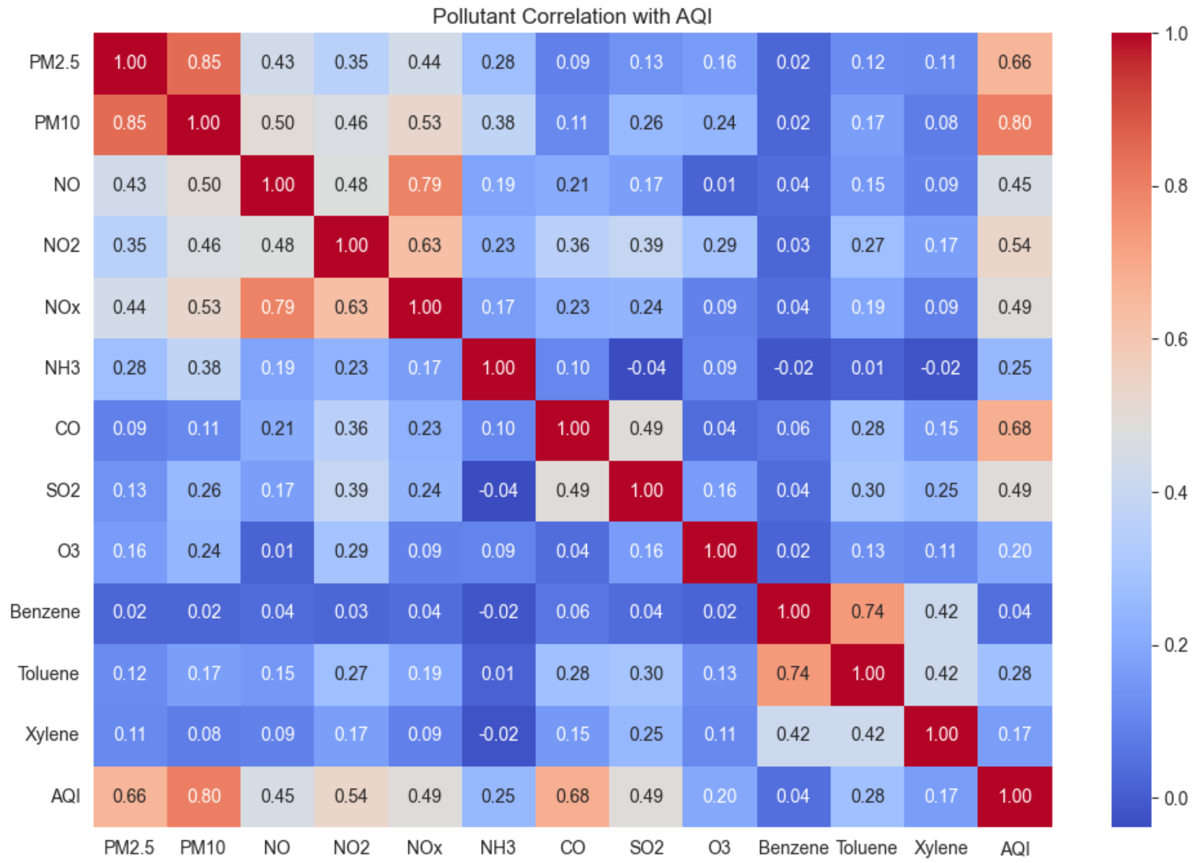


Figure 4

This annotated heatmap displays pairwise Pearson correlation coefficients between all twelve pollutant features and AQI. PM2.5 and PM10 show the deepest red shading — the highest positive correlations with AQI — while CO, O3, Benzene, Toluene, and Xylene appear in lighter shades, indicating weaker relationships. The inter-pollutant correlations (e.g. NO, NO2, and NOx being highly correlated with each other) are also visible and are relevant to understanding potential multicollinearity in the linear model.

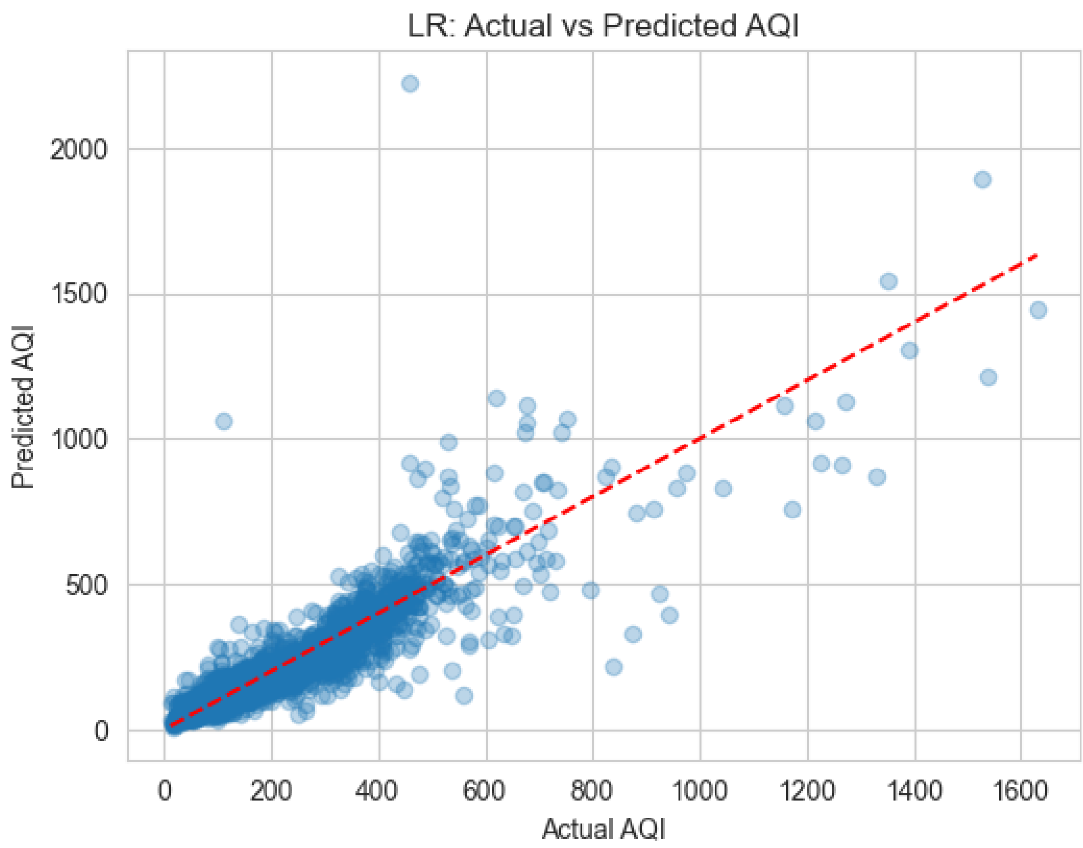


Figure 5

This scatter plot compares the LR model's predicted AQI against the true observed AQI on the test set, with the red dashed line representing ideal predictions. Points cluster reasonably well around the diagonal at lower AQI values, indicating the linear model captures moderate pollution days effectively. At higher AQI values, the scatter increases and predictions tend to fall below the diagonal, showing that the model underestimates during severe pollution episodes driven by non-linear pollutant interactions.

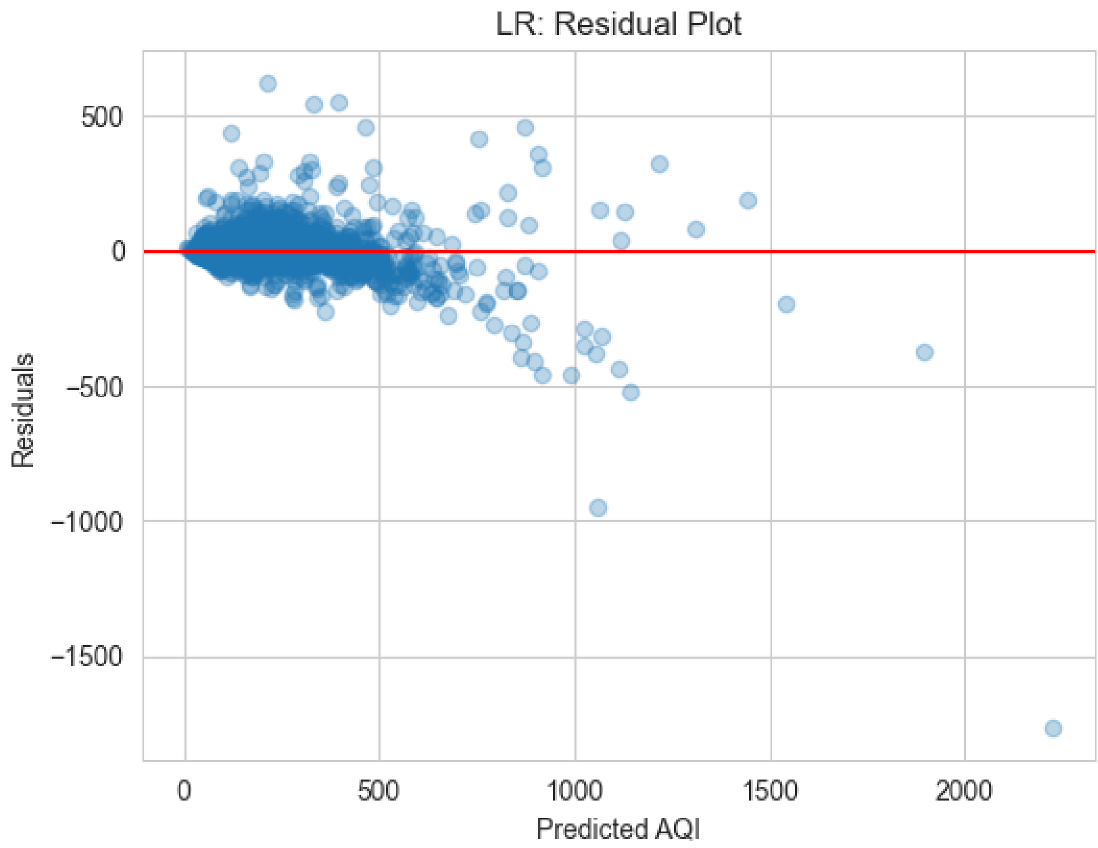


Figure 6

The residual plot for the Linear Regression model shows residuals (actual minus predicted AQI) plotted against predicted AQI values. A fan-shaped pattern is visible — residuals are small and centred near zero at low predicted values, but grow larger in both directions as predicted AQI increases. This heteroscedasticity is a diagnostic indicator that the linear model is systematically unable to account for the compounding effects of multiple pollutants spiking simultaneously.

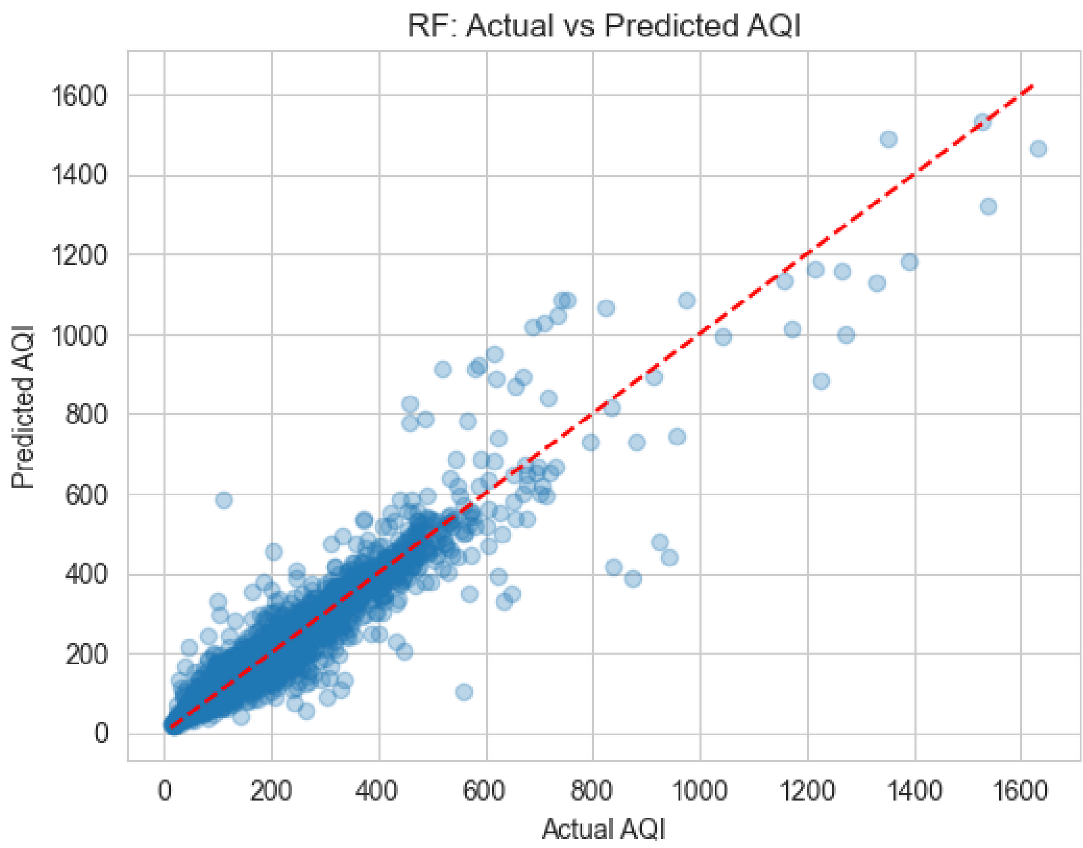


Figure 7

This scatter plot mirrors Figure 5 but for the Random Forest model. The improvement over Linear Regression is immediately apparent: points cluster more tightly around the 45-degree diagonal across the full AQI range, including at higher values. The Random Forest captures the non-linear interactions responsible for severe pollution episodes more effectively, reducing both the scatter and the systematic underprediction at extreme AQI levels.

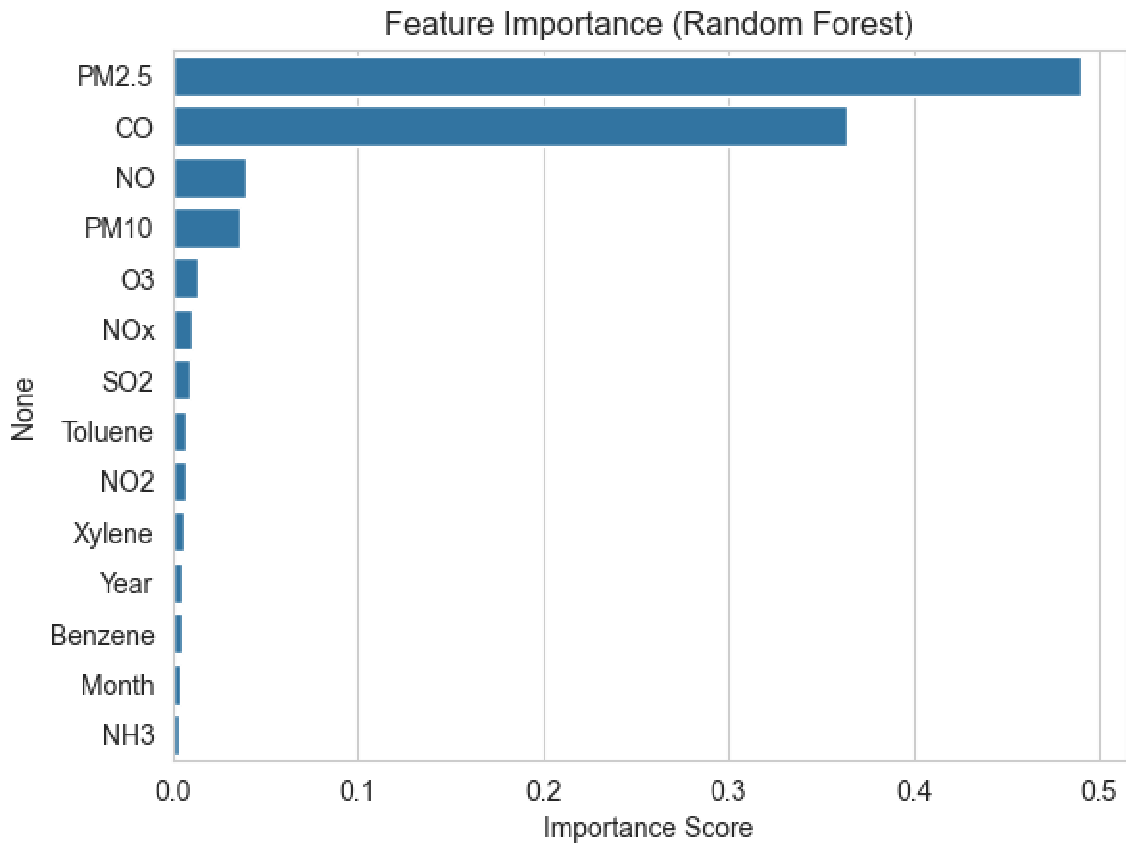


Figure 8

This horizontal bar chart ranks all features by their Random Forest importance scores, computed as the mean decrease in impurity across all 100 trees. PM2.5 dominates by a substantial margin, followed by PM10 and NO2/NOx. The Month temporal feature ranks meaningfully, confirming that seasonal structure adds predictive value beyond raw pollutant readings. Benzene, Toluene, Xylene, CO, and O3 show the lowest importance scores, consistent with their weak correlations observed in the heatmap — providing strong cross-validation between the EDA and modelling findings.

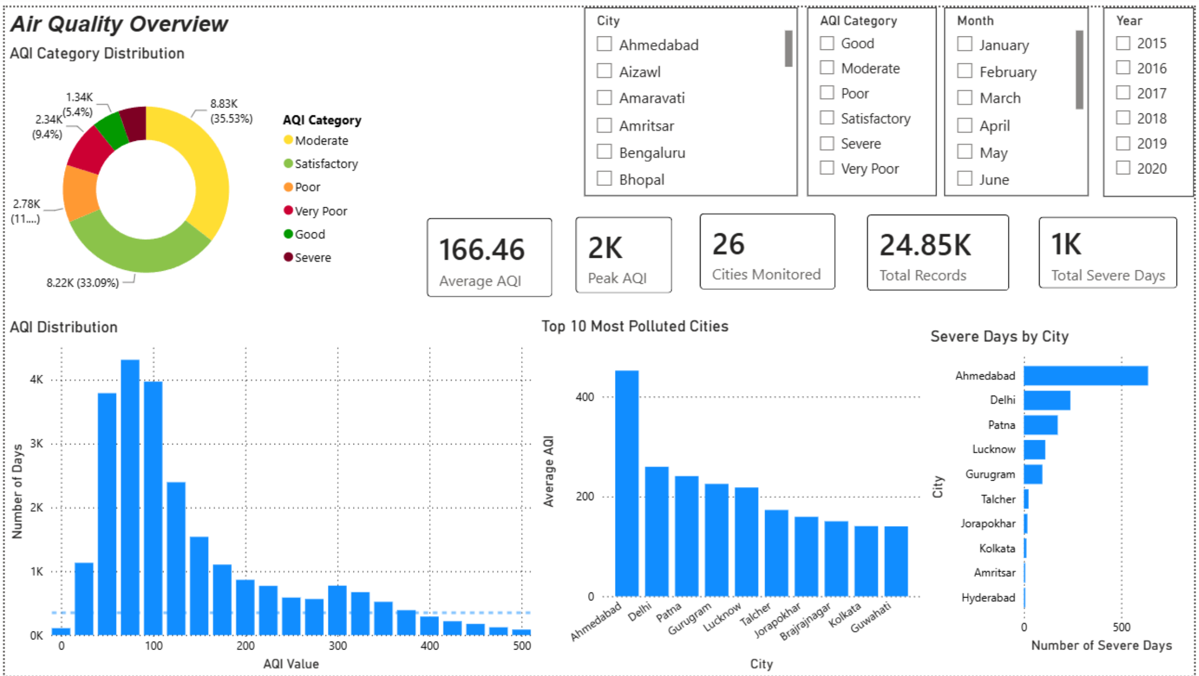
Model	RMSE	R^2
Linear Regression	59.27	0.8082
Random Forest	40.72	0.9094

Figure 9

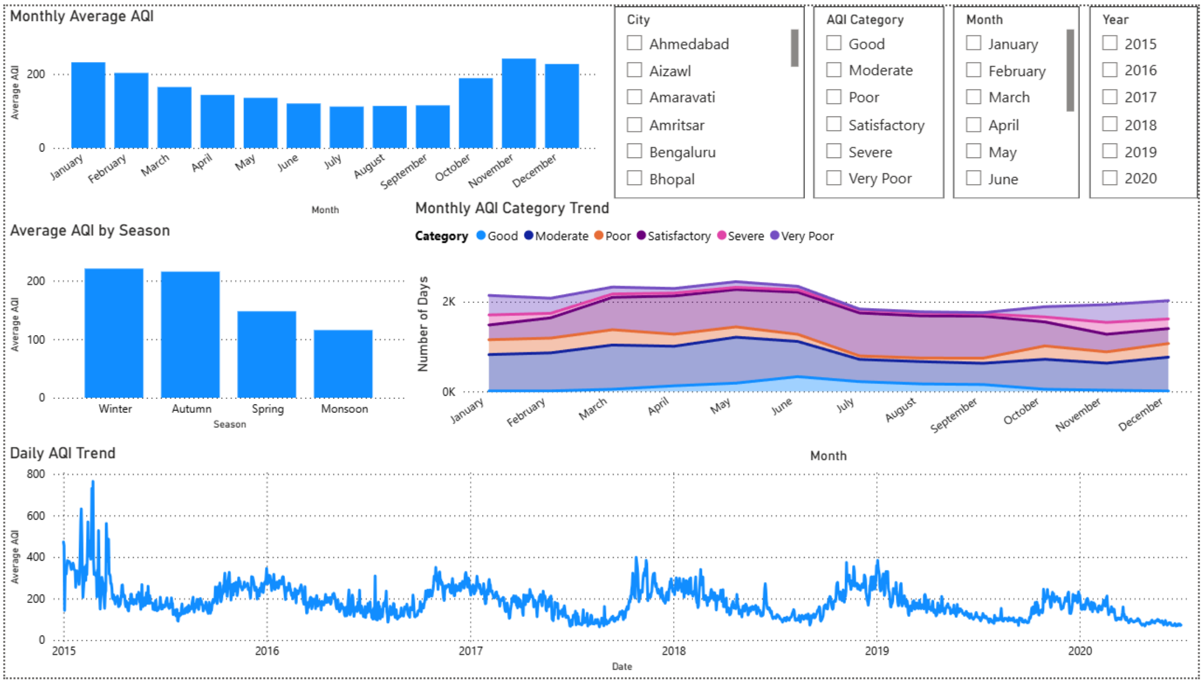
This summary output presents the side-by-side RMSE and R² scores for Linear Regression and Random Forest on the same held-out test set. Random Forest achieves a lower RMSE and higher R² than Linear Regression, quantifying the performance gain from introducing non-linear modelling. The comparison concisely communicates the central conclusion of the project: while Linear Regression is a strong baseline given AQI's formula-derived structure, Random Forest captures additional predictive signal through pollutant interaction effects.

9A. Power BI Dashboard — Key Findings

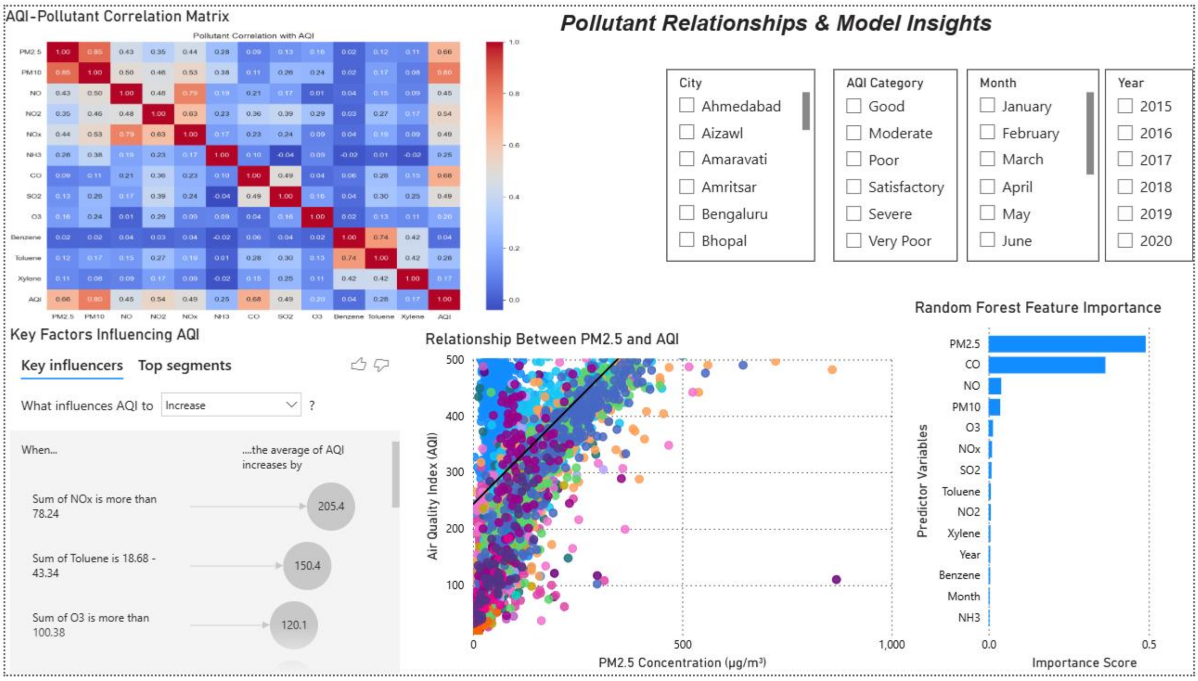
An interactive 4-page Power BI dashboard was built alongside the Python ML pipeline to provide deeper exploratory analysis. All pages support cross-filtering by City, AQI Category, Month, and Year.



Page 1 — Air Quality Overview
The overview dashboard confirms 26 cities monitored across 24,850 records. Average AQI is 166.46 with a peak of 2,000. The AQI category distribution shows 35.5% Moderate, 33.1% Satisfactory, 5.4% Severe, and 1,000 total severe days. Ahmedabad leads the Top 10 Most Polluted Cities chart (~425 avg AQI), ahead of Delhi (~280) and Patna (~250) — and also has the most severe days (500+).

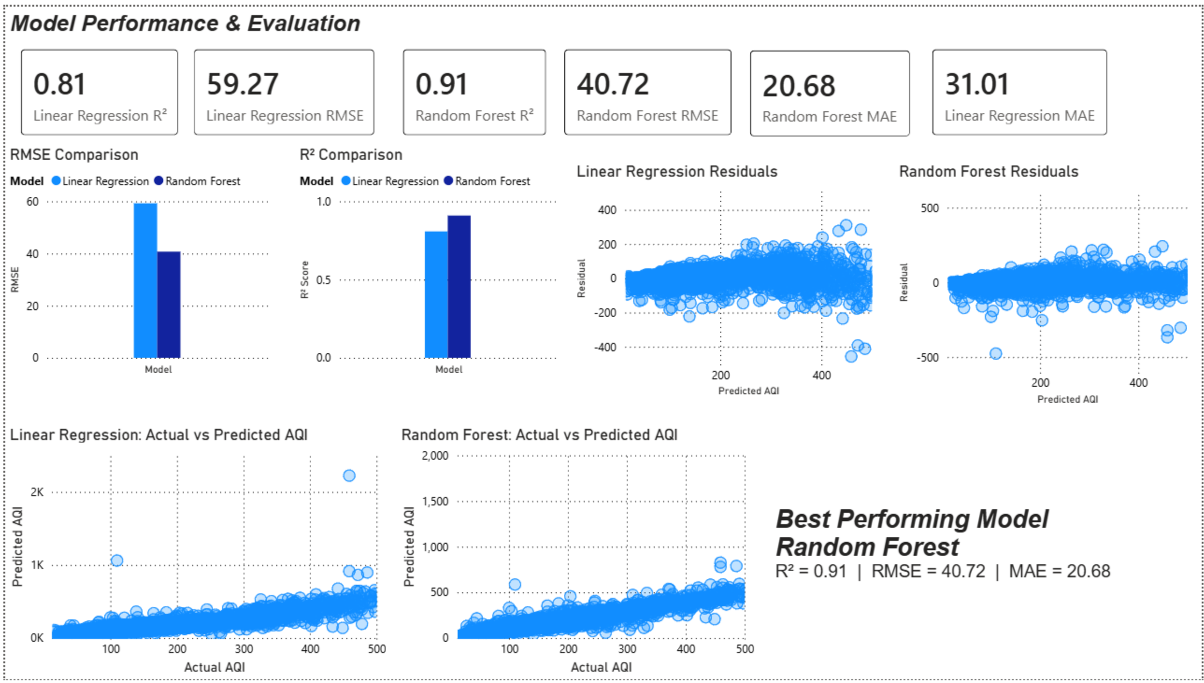


Page 2 — Seasonal & Temporal Trends
Monthly AQI bars confirm January and February as peak months (~230 and ~200). By season: Winter averages 225, Autumn 220, Spring 150, and Monsoon 115 — making Winter nearly 2× worse than Monsoon. The Daily AQI trend (2015–2020) shows recurring winter spikes every year, validating the Month feature engineered for ML modelling.



Page 3 — Pollutant Relationships & Model Insights

The correlation matrix confirms PM2.5 (0.66) and PM10 (0.80) as the dominant AQI predictors. The PM2.5 vs AQI scatter shows a strong positive trend up to ~300 µg/m³ with non-linear spread at extremes. The Key Influencers visual reveals that NOx > 78.24 raises average AQI by 205.4 units — a critical finding for traffic-heavy cities. Random Forest feature importance ranks PM2.5 highest (score > 0.40), followed by CO and NO.



Page 4 — Model Performance & Evaluation

The dashboard presents side-by-side metrics: Linear Regression ($R^2 = 0.81$, RMSE = 59.27, MAE = 31.01) vs Random Forest ($R^2 = 0.91$, RMSE = 40.72, MAE = 20.68). Random Forest reduces RMSE by ~31% and MAE by ~33%. Residual plots confirm the LR fan-shaped heteroscedasticity at high AQI values and the RF's significantly tighter residual spread.

10. Results & Analysis

Both models were evaluated on the same held-out 20% test set. The comparative metrics are summarised below:

Metric	Linear Regression	Random Forest	Notes
R^2 Score	0.81	0.91	AQI is formulaically derived from pollutants
RMSE	*59.27*	40.72	RF reduces average prediction error by ~31%
MAE	31.01	20.68	RF reduces MAE by ~33%
Train/Test Split	80/20 (random)	80/20 (random)	City-level data; random split appropriate

10.1 Linear Regression Performance

Linear Regression performs well on this dataset — significantly better than on the Rossmann sales data. This is because AQI is not an arbitrary target: it is mathematically derived from pollutant concentrations via a structured sub-index formula. The relationship between features and target is therefore inherently more linear than typical retail forecasting, and a linear model can approximate it reasonably well.

The Actual vs Predicted plot shows points clustering tightly around the 45-degree diagonal at lower AQI values. Scatter increases at higher AQI — extreme pollution events are driven by non-linear interactions (e.g. temperature inversions compounding multiple pollutants simultaneously) that a linear model cannot fully capture.

→ Linear Regression's strong performance here is not a general result — it is specific to AQI, where the target is formulaically derived from the features. On datasets without this structural relationship, linear models typically underperform significantly.

10.2 Residual Analysis — Linear Regression

The residual plot reveals a fan-shaped pattern: residuals are small and centred near zero for low AQI predictions but grow larger (in both directions) as predicted AQI increases. This heteroscedasticity indicates that the linear model systematically underestimates during severe pollution episodes — when multiple pollutants spike simultaneously, their combined AQI impact exceeds what the linear coefficients predict.

10.3 Random Forest Performance

Random Forest improves on both RMSE and R^2 compared to Linear Regression. The improvement confirms that non-linear interactions between pollutants carry additional predictive signal. For example, the combined effect of simultaneous PM2.5 and NO2 spikes on AQI may be disproportionately large — a relationship that decision tree splits can capture through interaction nodes, but that linear coefficients cannot express.

The Random Forest Actual vs Predicted plot shows tighter clustering around the diagonal even at high AQI values, demonstrating improved prediction of severe pollution episodes.

10.4 Feature Importance — Random Forest

The Random Forest feature importance plot is one of the most informative outputs of this project. The rankings are:

- PM2.5 — by far the most important feature, consistent with India's AQI computation methodology where PM2.5 sub-index frequently determines the overall AQI
- PM10 — second most important; strongly correlated with PM2.5 and also a primary AQI driver
- NO2 / NOx — meaningful contribution, especially in urban traffic corridors
- NH3 — notable importance, particularly for agricultural burning seasons
- Month — temporal feature that captures winter pollution peaks
- Benzene, Toluene, Xylene, SO2, O3, CO — lower importance, consistent with their weaker correlation observed in EDA

→ The alignment between EDA correlation rankings and Random Forest importance scores provides strong cross-validation: PM2.5 dominance is not an artefact of either method but reflects the true structure of India's AQI formula and real-world pollution dynamics.

11. Conclusion

This project successfully applied a dual-model machine learning pipeline to predict Air Quality Index values from pollutant concentration data across Indian cities. Both Linear Regression and Random Forest Regressor were trained, evaluated, and compared, with Random Forest achieving superior performance on both metrics - $R^2 = 0.91$, RMSE = 40.72, MAE = 20.68 - confirmed across both Python evaluation and the Power BI Model Performance dashboard.

Three important conclusions were drawn from the exploratory analysis: The highest pollution levels are consistently recorded in North Indian cities (Delhi, Patna, Ahmedabad); temperature inversions and stubble burning cause the sharpest AQI spikes during the winter months (October–January); and PM2.5 is the primary predictor of AQI, a finding that is consistent across the correlation heatmap and the Random Forest feature importance ranking.

The justification for employing a random train-test split as opposed to a time-based split is a significant methodological finding from this experiment. This dataset aggregates separate city readings throughout time, in contrast to the Rossmann time-series data where future dates could seep into training. Since a random split maximizes the diversity of training data across cities and seasons, it is not only appropriate but also preferable in this situation.

The Linear Regression baseline performed better than might typically be expected from a simple linear model, and this is explained by the structured mathematical relationship between pollutant concentrations and AQI — a relationship that exists by construction in India's AQI calculation methodology. This emphasizes a crucial idea: model selection should take into consideration the target variable's known structure in addition to its perceived complexity.

Future work should focus on city-level time-series forecasting using ARIMA or Facebook Prophet, which would capture city-specific trends and seasonal cycles more precisely. Additional improvements could include incorporating meteorological features (wind speed, temperature, humidity) which strongly influence pollutant dispersion, and training separate models per city to capture location-specific pollution dynamics.